WORKED SOLUTION

STEP III 2023, Question 11

Probability and Statistics • Conditional Expectation

STEP III 2023, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The opening series splits into two familiar exponential series once we write $k+1 = k+1$ and cancel the k against $k!$. In part (i) the distribution of D involves a sum over every value of Y that could have produced $D = d$; swapping the order of the double sum makes the inner sum the easy $1+2+\cdots+k$, and the factor $\frac{1}{2}(k+1)$ that appears is exactly what the opening series handles. Part (ii) is then a matter of conditioning on the die.

Solution

The series

For $k \geq 1$ we have $\dfrac{k}{k!} = \dfrac{1}{(k-1)!}$, so, splitting the numerator,

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{k+1}{k!} x^k &= \sum_{k=1}^{\infty} \frac{x^k}{(k-1)!} + \sum_{k=1}^{\infty} \frac{x^k}{k!} \\ &= x \sum_{j=0}^{\infty} \frac{x^j}{j!} + \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} - 1 \right) \\ &= x\mathrm{e}^x + \mathrm{e}^x - 1 = (x+1)\mathrm{e}^x - 1,\end{aligned}$$

using $j = k-1$ in the first sum. Both series converge for every real x, so splitting is legitimate. This is the required identity.

Part (i)(a)

$D = 0$ exactly when $Y = 0$ (if $Y = k \geq 1$, the die shows at least 1), so

$$\mathrm{P}(D = 0) = \mathrm{P}(Y = 0) = \mathrm{e}^{-n}.$$

Part (i)(b)

Fix $d \geq 1$. The die can show d only if it has at least d sides, that is, only if $Y = k$ with $k \geq d$; and given $Y = k \geq d$, the fair die shows d with probability $\frac{1}{k}$. By the law of total probability,

$$\mathrm{P}(D = d) = \sum_{k=d}^{\infty} \mathrm{P}(D = d \mid Y = k)\,\mathrm{P}(Y = k) = \sum_{k=d}^{\infty} \frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n}.$$

By definition $\mathrm{E}(D) = \sum_{d=0}^{\infty} d\,\mathrm{P}(D = d)$, and the $d = 0$ term is zero, so

$$\mathrm{E}(D) = \sum_{d=1}^{\infty} \left[d \sum_{k=d}^{\infty} \left(\frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \right) \right],$$

as required.

Ask yourself How do we turn a sum over d outside and $k \geq d$ inside into a sum over k outside?

The double sum runs over all pairs (d, k) of integers with $1 \leq d \leq k$. Every term $\frac{d}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n}$ is non-negative, so we may add the terms in any order without changing the total. Grouping by k instead of by d: for each $k \geq 1$, d runs from 1 to k. Hence

$$\mathrm{E}(D) = \sum_{k=1}^{\infty} \left(\frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \sum_{d=1}^{k} d \right),$$

which is the second required form.

Part (i)(c)

Using $\sum_{d=1}^{k} d = \frac{1}{2}k(k+1)$, the factor $\frac{1}{k}$ cancels:

$$\begin{aligned} \mathrm{E}(D) &= \sum_{k=1}^{\infty} \frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \cdot \frac{k(k+1)}{2} = \frac{\mathrm{e}^{-n}}{2} \sum_{k=1}^{\infty} \frac{k+1}{k!} n^k \\ &= \frac{\mathrm{e}^{-n}}{2}\left((n+1)\mathrm{e}^{n} - 1 \right) = \tfrac{1}{2}\left(n + 1 - \mathrm{e}^{-n} \right), \end{aligned}$$

by the opening identity with $x = n$. This is the required result.

Part (ii)(a)

Let K be the number shown on the n-sided die, so $\mathrm{P}(K = k) = \frac{1}{n}$ for $k = 1, \ldots, n$, and given $K = k$ we have $Z = X_k$, which is Poisson with mean k. So $\mathrm{P}(Z = 0 \mid K = k) = \mathrm{e}^{-k}$, and by the law of total probability

$$\mathrm{P}(Z = 0) = \frac{1}{n} \sum_{k=1}^{n} \mathrm{e}^{-k}.$$

This is a geometric series with first term e^{-1} and ratio e^{-1}:

$$\mathrm{P}(Z=0) = \frac{1}{n}\cdot\frac{\mathrm{e}^{-1}\left(1-\mathrm{e}^{-n}\right)}{1-\mathrm{e}^{-1}} = \frac{1-\mathrm{e}^{-n}}{n(\mathrm{e}-1)},$$

multiplying top and bottom by e.

Part (ii)(b)

Conditioning on the die again, $\mathrm{E}(Z \mid K=k) = \mathrm{E}(X_k) = k$, so

$$\mathrm{E}(Z) = \sum_{k=1}^{n} \mathrm{E}(Z \mid K=k)\,\mathrm{P}(K=k) = \frac{1}{n}\sum_{k=1}^{n} k = \frac{1}{n}\cdot\frac{n(n+1)}{2} = \frac{n+1}{2}.$$

Comparing with part (i)(c),

$$\mathrm{E}(Z) - \mathrm{E}(D) = \frac{n+1}{2} - \frac{n+1-\mathrm{e}^{-n}}{2} = \tfrac{1}{2}\mathrm{e}^{-n} > 0,$$

so $\mathrm{E}(Z) > \mathrm{E}(D)$, as required.

Interestingly, without the zero case the two would agree: a fair die with a Poisson number Y of faces shows $\frac{1}{2}(Y+1)$ on average, which has mean $\frac{1}{2}(n+1)$. The only loss is that $Y=0$ gives $D=0$ rather than the $\frac{1}{2}$ that the formula $\frac{1}{2}(Y+1)$ would give, and that happens with probability e^{-n}, costing exactly $\frac{1}{2}\mathrm{e}^{-n}$.

Checks

- With $x=0$ the opening identity reads $0 = 1\cdot 1 - 1$, and with $x=1$ it gives $\sum_{k\geq 1}\frac{k+1}{k!} = 2\mathrm{e}-1 \approx 4.4366$, which the partial sums confirm.
- For $n=3$: $\mathrm{E}(D) = \frac{1}{2}(4-\mathrm{e}^{-3}) \approx 1.9751$, $\mathrm{E}(Z) = 2$ and $\mathrm{P}(Z=0) \approx 0.1843$. A computer simulation of 200 000 runs of each experiment gave 1.970, 2.002 and 0.185.
- For $n=1$, the die in part (ii) always shows 1, so $\mathrm{P}(Z=0) = \mathrm{e}^{-1}$; the formula gives $\frac{1-\mathrm{e}^{-1}}{\mathrm{e}-1} = \mathrm{e}^{-1}$.

Common slips

- Starting the inner sum in (i)(b) at $k=1$ rather than $k=d$: a k-sided die cannot show a number bigger than k.
- Forgetting the -1 in the opening identity, which comes from the missing $k=0$ term of e^{x}; it is the source of the e^{-n} in (i)(c).

Beyond STEP

Both parts are instances of the tower law $\mathrm{E}(Z) = \mathrm{E}\big(\mathrm{E}(Z \mid K)\big)$: in part (i) it gives $\mathrm{E}(D) = \mathrm{E}\big(\frac{1}{2}(Y+1)\mathbf{1}_{Y\geq 1}\big)$ in one line, and the swap of summation order we did by

hand is exactly what that law hides. The freedom to rearrange a double series of non-negative terms is a theorem (Tonelli's, in its general form) met in first-year analysis. Grimmett and Welsh, *Probability: An Introduction*, treats conditional expectation and the compound distributions built from it.

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